Exploratory Data Analysis (EDA)

Developer Survey Data Analysis

AI Adoption, Salary and Developer Insights

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1 Introduction

1.1 Background and Context

Artificial Intelligence (AI) tools have rapidly become integrated into modern software development workflows. From automated code generation to debugging assistance and documentation support, AI-powered systems are transforming how developers design, implement, and maintain software solutions.

In recent years, tools such as ChatGPT, GitHub Copilot, and other intelligent assistants have gained widespread adoption among developers worldwide. This technological shift raises important questions regarding productivity, income, and job satisfaction within the software engineering profession.

1.2 Problem Statement

While AI adoption is increasing, its actual impact on developers' professional outcomes remains a subject of debate. Does the use of AI tools influence salary levels? Is income primarily driven by years of professional experience? Does AI improve job satisfaction? Understanding these relationships is essential for evaluating the evolving role of AI in the technology labor market.

1.3 Objectives of the Study

The main objective of this study is to conduct an Exploratory Data Analysis (EDA) on developer survey data in order to examine the relationship between:

- AI tool adoption among developers,
- Salary distribution,
- Professional experience,
- Job satisfaction levels.

This study aims to identify dominant developer profiles, explore salary determinants, and assess whether AI usage is associated with measurable professional advantages.

1.4 Research Questions

This report seeks to answer the following research questions:

- How widely are AI tools adopted among developers?
- Do developers who use AI tools earn higher salaries?
- Is salary primarily influenced by professional experience?
- Does AI usage correlate with higher job satisfaction?

1.5 Structure of the Report

The remainder of this report is organized as follows: Section 2 presents the dataset and variables used in the analysis. Section 3 describes the data preprocessing and methodology. Sections 4 and 5 provide descriptive and bivariate analyses. Section 6 examines correlations between key variables. Finally, the report concludes with a discussion of findings and implications.

2 Data Description

2.1 Dataset Source

This study is based on the Stack Overflow Annual Developer Survey 2024 dataset, publicly available on Kaggle. The survey collected responses from over 65,000 developers worldwide, covering topics such as employment, salary, education, programming technologies, and Artificial Intelligence usage.

The original dataset contains 114 variables capturing demographic, professional, and technological aspects of developers.

2.2 Sample Overview

The dataset includes approximately 65,437 observations, each representing one survey respondent.

For the purpose of this analysis, only a subset of relevant variables was selected to address the research questions regarding salary, professional experience, AI adoption, and job satisfaction.

2.3 Selected Variables

The following key variables were retained:

Quantitative Variables:

- ConvertedCompYearly: Annual salary (in USD).
- WorkExp: Years of professional experience.
- JobSat: Job satisfaction level (treated as a numerical scale for analytical purposes).

Qualitative Variables:

- Country: Respondent's country.
- DevType: Developer role (multi-select).
- LanguageHaveWorkedWith: Programming languages used.
- AIToolCurrently Using: AI tools currently used.
- AISearchDevHaveWorkedWith: AI-related tools or search systems used.

2.4 Derived Variables

To improve interpretability and analytical consistency, additional variables were created:

- Salary_log10: Logarithmic transformation of salary to reduce skewness.
- WorkExp_num: Numeric conversion of work experience.
- DevType_main: Primary developer role extracted from multi-select responses.
- UsesAI_current: Binary indicator of current AI tool usage.

2.5 Data Characteristics

The dataset combines categorical (nominal and ordinal) variables with continuous numerical variables. Salary data exhibits strong positive skewness, which justifies the application of logarithmic transformation during analysis.

Given the international scope of the survey, geographic distribution must be considered when interpreting salary comparisons.

3 Methodology

This section describes the data preparation process and analytical steps undertaken to ensure consistency, reliability, and interpretability of the results. Although JobSat is inherently ordinal, it was treated as a numerical variable to facilitate statistical analysis.

3.1 Data Loading

Dataset Source

- Stack Overflow Developer Survey 2024 (Kaggle).
- 65,437 observations and 114 variables.
- Covers demographic, professional, and technological attributes.

Data Import Process

```
import pandas as pd
import numpy as np

# Load dataset
path = "/kaggle/input/stack-overflow-annual-developer-survey-2024/survey_results_public.csv"
df_raw = pd.read_csv(path, low_memory=False)

# Check dataset dimensions
print(df_raw.shape)
df_raw.head()
```

3.2 Variable Selection

Quantitative Variables

- ConvertedCompYearly
- WorkExp
- JobSat (treated as numerical scale)

Categorical Variables

- Country
- DevType
- LanguageHaveWorkedWith
- AIToolCurrentlyUsing
- AISearchDevHaveWorkedWith

3.3 Data Cleaning

To ensure data reliability and analytical validity, several preprocessing steps were applied.

First, salary and professional experience variables were converted to numeric format using safe coercion. Invalid or non-positive salary values were removed to prevent distortion of statistical measures.

Extreme salary outliers (top 1%) were filtered using quantile-based trimming. Job satisfaction values were restricted to the valid response range.

Categorical text variables were standardized by trimming whitespace and converting formats to ensure consistency. Multi-select responses (e.g., developer roles and AI tools) were cleaned to facilitate structured analysis.

Preprocessing Steps

- Numeric conversion with safe coercion
- Removal of invalid salary values
- Outlier filtering (Top 1%)
- Text standardization
- Handling missing values

```
# Remove duplicates
df = df_raw.drop_duplicates()

# Convert to numeric
df["ConvertedCompYearly"] = pd.to_numeric(df["ConvertedCompYearly"], errors="coerce")
df["WorkExp"] = pd.to_numeric(df["WorkExp"], errors="coerce")
df["JobSat"] = pd.to_numeric(df["JobSat"], errors="coerce")

# Remove missing key values
df = df.dropna(subset=["ConvertedCompYearly", "WorkExp"])

# Keep only positive salaries
df = df[df["ConvertedCompYearly"] > 0]

# Remove extreme salary outliers (top 1%)
upper_salary = df["ConvertedCompYearly"].quantile(0.99)
df = df[df["ConvertedCompYearly"] <= upper_salary]

# Valid job satisfaction range
df = df[df["JobSat"].between(0, 9)]

# Standardize text columns
for col in ["Country", "DevType", "Employment"]:
    if col in df.columns:
        df[col] = df[col].astype(str).str.strip()
```

3.4 Feature Engineering

To enhance analytical interpretability and reduce statistical distortions, additional features were engineered from the raw variables.

A logarithmic transformation was applied to annual salary to address strong positive skewness and mitigate the influence of extreme high-income values.

Professional experience was retained in numeric form to ensure compatibility with regression-based and correlation analyses.

For categorical multi-select variables, structured transformations were applied to extract primary developer roles and facilitate AI usage analysis. **Key Transformations**

- Log transformation of salary
- Numeric encoding of experience
- Extraction of primary developer role
- Binary AI usage indicator

```
# Log transformation of salary
df["Salary_log10"] = np.log10(df["ConvertedCompYearly"])

# Ensure experience is numeric
df["WorkExp_num"] = pd.to_numeric(df["WorkExp"], errors="coerce")

# Extract primary developer role
df["DevType_main"] = (
    df["DevType"]
    .astype(str)
    .str.split(";")
    .str[0]
    .str.strip()
)

# AI usage flag (0/1)
df["UsesAI"] = df["AIToolCurrently␣Using"].notna().astype(int)
```

3.5 Analytical Approach

The analytical strategy was structured in three sequential stages to ensure a comprehensive exploration of the dataset.

1. Descriptive Analysis: Initial exploration involved examining summary statistics and distributional properties of both quantitative and categorical variables. This step provided insights into central tendencies, variability, and distribution shapes, particularly addressing salary skewness.

2. Bivariate Analysis: Pairwise relationships between key variables were investigated using scatter plots, boxplots, and grouped comparisons. This stage focused on understanding associations between professional experience and salary, AI usage and income levels, and role specialization effects.

3. Correlation Analysis: Pearson correlation coefficients were computed to quantify the strength and direction of linear associations among numerical variables. A heatmap visualization was used to summarize correlation patterns.

All visualizations were generated using the Matplotlib and Seaborn libraries in Python.

3.6 Methodological Limitations

Several methodological limitations should be acknowledged.

First, the dataset is based on self-reported survey responses, which may introduce reporting bias and measurement inaccuracies.

Second, the analysis is cross-sectional, limiting the ability to establish causal relationships between AI adoption and salary outcomes. Observed associations reflect correlation rather than causation.

Third, salary differences may be influenced by unobserved confounding variables such as education level, company size, industry sector, or geographic economic conditions.

Finally, simplification of multi-select variables (e.g., extraction of a primary developer role) may reduce nuanced information contained in the original responses.

4 Descriptive Analysis

4.1 Qualitative Variables

This section examines the distribution of categorical variables in the dataset, including geographic location, developer roles, programming languages, AI adoption, AI use cases, and job satisfaction levels. The objective is to identify dominant patterns and overall trends within the developer population.

4.1.1 Geographic Distribution

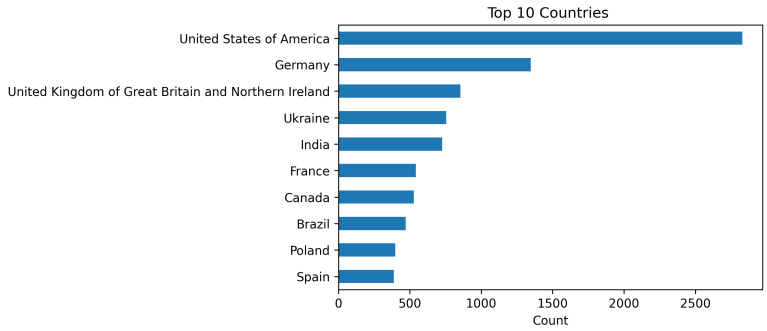

Figure 1: Top 10 Countries by Number of Respondents

The dataset is heavily dominated by respondents from the United States, which accounts for the largest proportion of participants by a substantial margin. Germany ranks second, followed by the United Kingdom and Ukraine. India, France, Canada, Brazil, Poland, and Spain also contribute notable shares of responses.

This geographic concentration suggests that salary comparisons and technological trends should be interpreted with awareness of regional economic differences. The dominance of North America and Europe may influence aggregate salary statistics and AI adoption rates.

4.1.2 Developer Roles

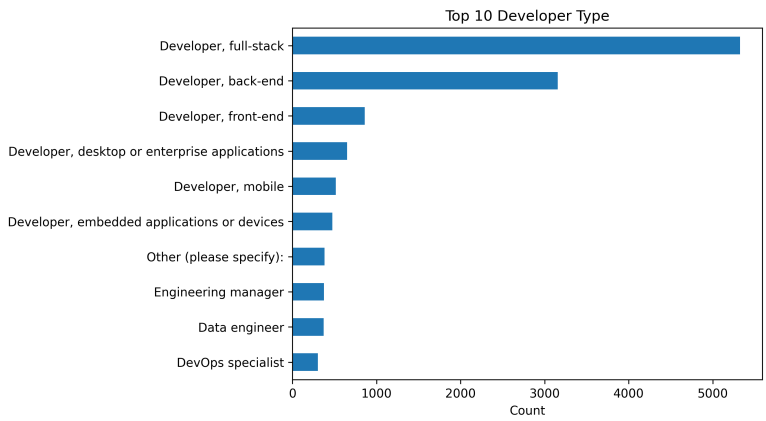

Figure 2: Top 10 Developer Roles

Full-stack developers represent the largest professional group in the dataset, followed by back-end developers. Front-end developers account for a smaller yet still substantial proportion of respondents.

Other roles, including mobile developers, embedded systems developers, data engineers, DevOps specialists, and engineering managers, appear with comparatively lower frequencies.

The prominence of full-stack and back-end roles may reflect the strong representation of web-oriented development within the survey sample. This distribution is important for subsequent salary comparisons, as role specialization is expected to influence income levels.

4.1.3 Programming Languages

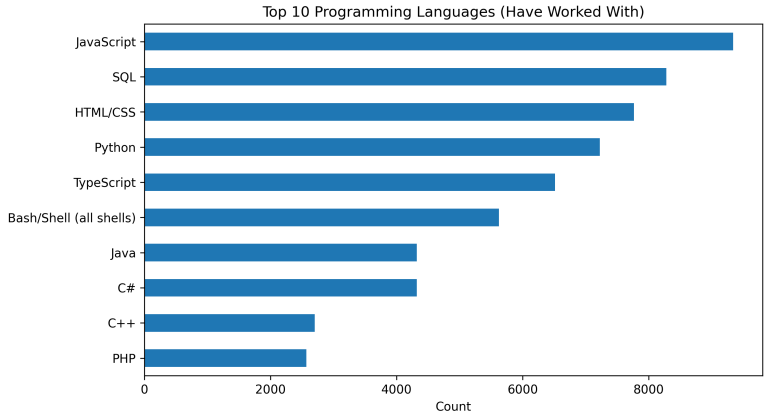

Figure 3: Top 10 Programming Languages Used

JavaScript is the most widely used programming language among respondents, followed by SQL and HTML/CSS. Python ranks fourth but remains highly prominent, indicating its continued growth across domains.

TypeScript, Bash/Shell, Java, and C# also maintain strong representation, while C++ and PHP complete the top ten.

The prevalence of JavaScript and web-related technologies confirms the central role of web development in modern software engineering.

4.1.4 AI Adoption Rate

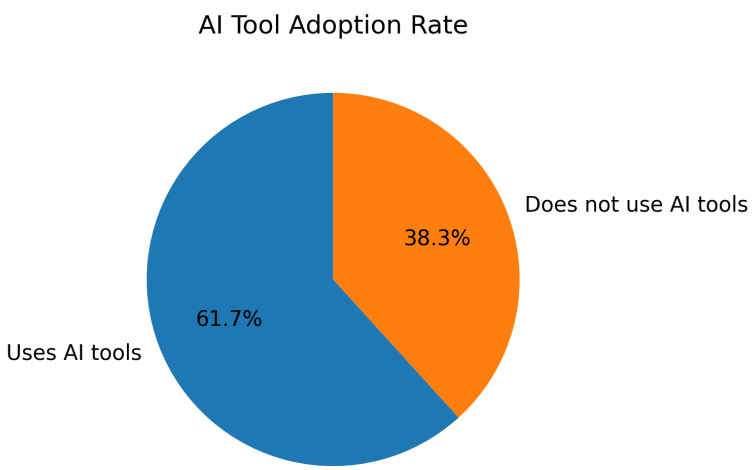

Figure 4: AI Tool Adoption Rate

Approximately 61.7% of respondents report using AI tools in their development workflow, while 38.3% do not use such tools.

This indicates that AI adoption has already reached mainstream levels among developers. The majority usage suggests that AI tools are increasingly integrated into daily software development practices.

4.1.5 AI Use Cases

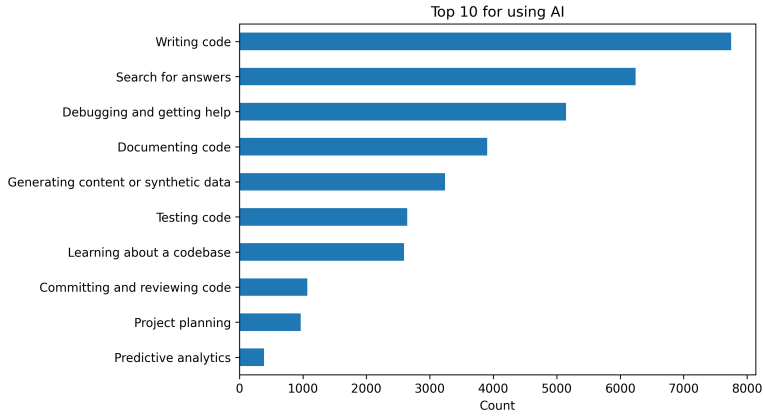

Figure 5: Top Use Cases for AI Tools

The most common use of AI tools is writing code, followed by searching for answers and debugging assistance. Documentation generation and content creation also appear as important use cases.

Less frequent applications include project planning and predictive analytics.

These results indicate that developers primarily rely on AI tools to enhance productivity in core programming tasks rather than high-level strategic decision-making.

4.1.6 Most Used AI Tools

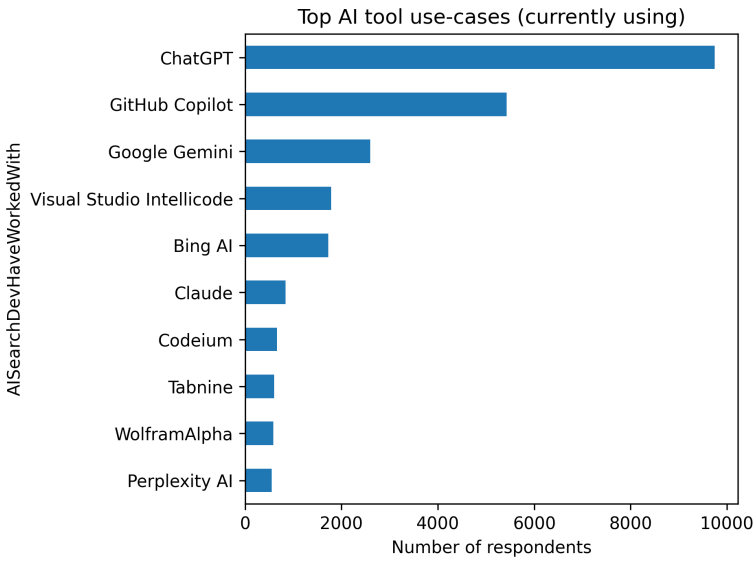

Figure 6: Top AI Tools Currently Used

ChatGPT is the most widely used AI tool among respondents, significantly outperforming other tools. GitHub Copilot follows as the second most used AI assistant.

Other tools such as Google Gemini, Bing AI, and Visual Studio IntelliCode show moderate adoption, while tools like Claude, Codeium, Tabnine, and Perplexity AI appear less frequently.

This distribution highlights the strong dominance of large language model-based assistants in modern development workflows.

4.2 Quantitative Variables

4.2.1 Job Satisfaction Distribution

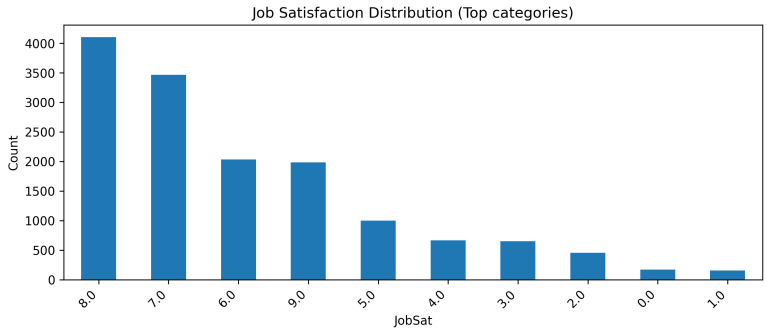

Figure 7: Job Satisfaction Distribution

Job satisfaction levels are generally skewed toward higher values. Ratings of 7 and 8 are the most common, indicating relatively high overall satisfaction among developers.

Lower satisfaction scores appear less frequently, suggesting that the majority of respondents report moderate to high professional satisfaction.

This distribution will be further examined in relation to AI usage in the bivariate analysis section.

4.2.2 Salary Distribution

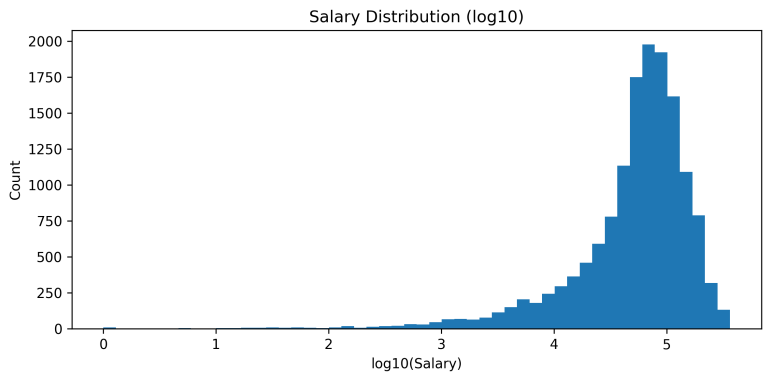

Figure 8: Log-Transformed Salary Distribution

The distribution of annual salary, after applying a base-10 logarithmic transformation, appears substantially more symmetric and closer to a normal distribution. This transformation was necessary due to the strong positive skewness observed in the raw salary data, where a small number of very high-income observations distorted the distribution.

Most respondents cluster around log10 values between approximately 4.5 and 5.2. Converting these values back to the original scale suggests that the majority of developers earn between roughly $30,000 and 160{,}000$ annually, indicating a concentration around mid-to-upper income levels.

The log transformation reduces the influence of extreme outliers and enables more reliable statistical comparisons in subsequent analyses.

4.2.3 Work Experience Distribution

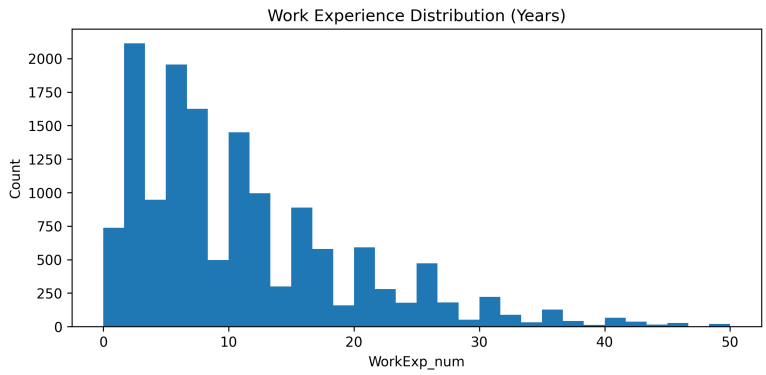

Figure 9: Work Experience Distribution (Years)

The distribution of professional experience is positively skewed, indicating that a large proportion of respondents have lower to mid-range levels of experience.

Most developers report between 2 and 15 years of professional experience. The frequency gradually declines as experience increases, with relatively few respondents exceeding 30 years.

This pattern likely reflects the demographic composition of the survey sample, which appears to include a substantial share of early- to mid-career professionals.

5 Bivariate Analysis

This section investigates relationships between key variables in order to identify meaningful patterns beyond simple distributional descriptions. Unlike the previous section, which focused on univariate characteristics, the present analysis explores potential associations between professional experience, salary, AI adoption, and job satisfaction.

5.1 Professional Experience vs Salary

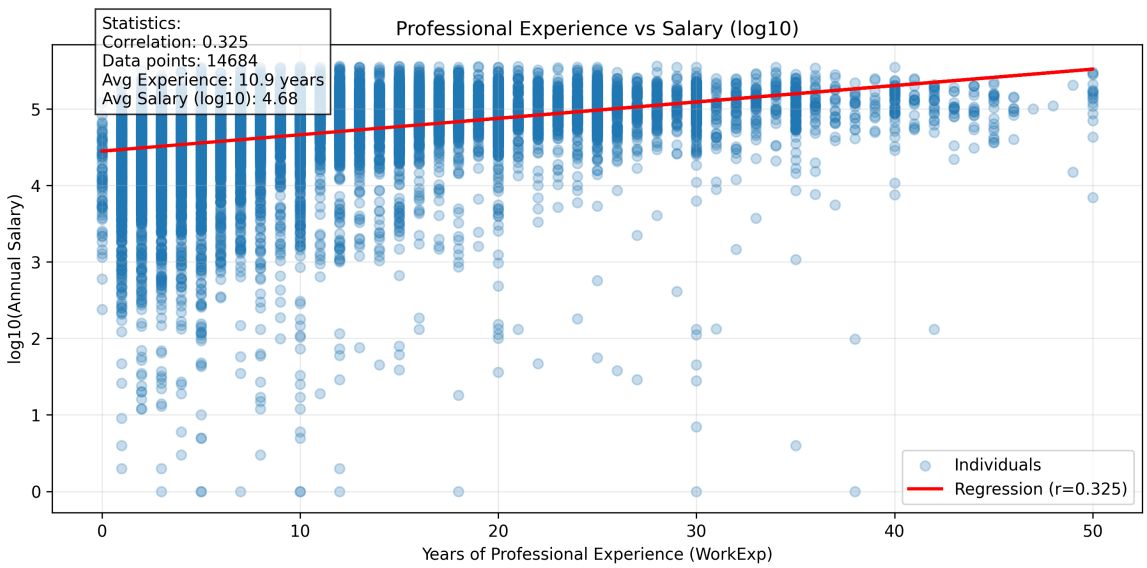

Figure 10: Professional Experience vs Log-Transformed Salary

The relationship between professional experience and salary reveals a clear positive trend. The regression line indicates that salary generally increases as years of experience grow. The reported Pearson correlation coefficient (r = 0.325) suggests a weak-to-moderate positive association between the two variables.

While statistically meaningful, the relationship is not strong, indicating that experience alone does not fully explain salary variation. The noticeable dispersion of points around the regression line highlights substantial heterogeneity within experience groups, suggesting the influence of additional factors such as geographic location, role specialization, and industry sector.

Overall, the upward slope aligns with human capital theory, where accumulated experience contributes positively to income, although other determinants clearly play a significant role.

5.2 Salary vs AI Usage

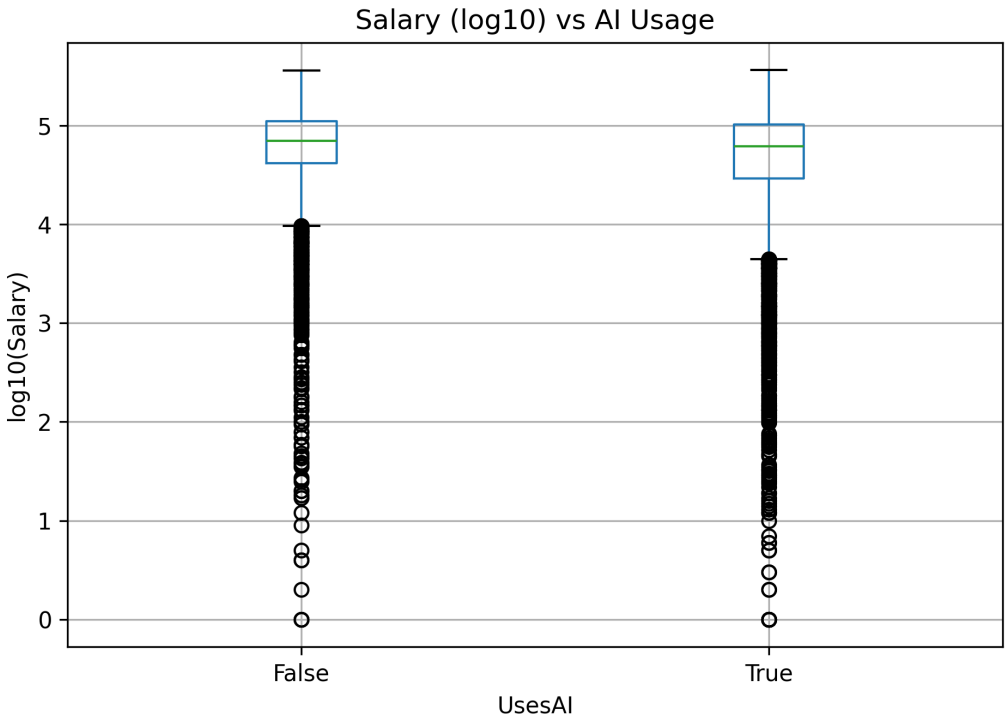

Figure 11: Log-Transformed Salary by AI Usage

The boxplot comparison between AI users and non-users indicates a slightly higher median salary among developers who report using AI tools. However, the difference between the two groups appears relatively small.

The interquartile ranges (IQRs) are largely comparable, and there is substantial overlap between the distributions. This suggests that while AI users tend to earn marginally higher salaries on average, AI usage alone does not appear to be a dominant driver of income differences.

Importantly, this analysis reflects association rather than causation. Higher-income developers may be more likely to adopt AI tools, or AI usage may correlate with roles that already command higher compensation.

5.3 Salary Differences Across Countries

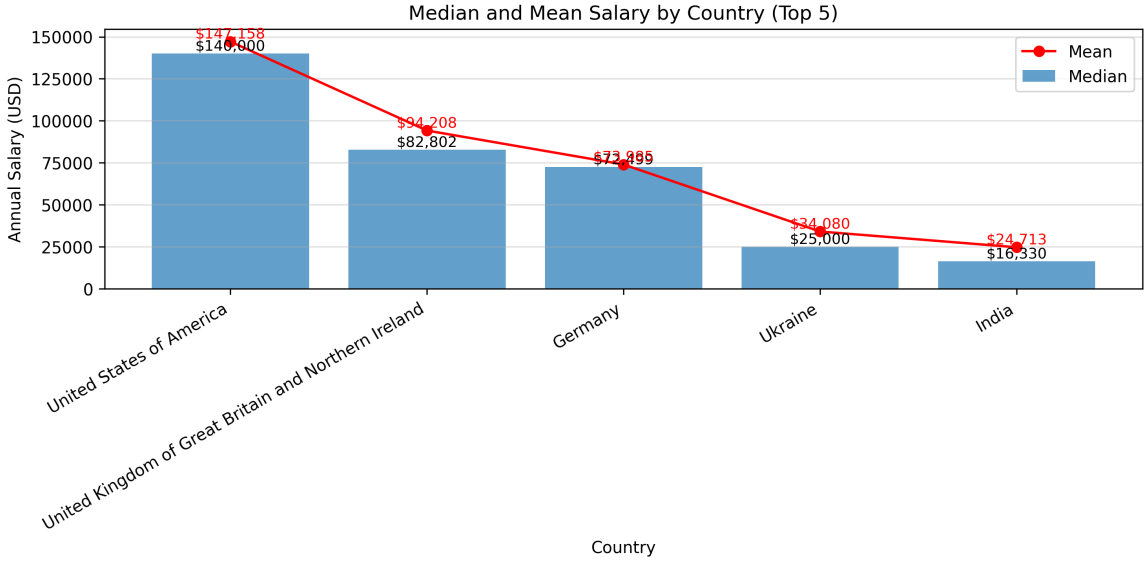

Figure 12: Median and Mean Salary by Country (Top 5)

Substantial cross-country salary disparities are evident. The United States exhibits the highest median and mean salary levels, considerably exceeding those observed in the other countries shown. The United Kingdom and Germany follow, while Ukraine and India display markedly lower compensation levels.

Across all countries, mean salaries are consistently higher than median salaries, indicating the presence of high-income outliers that elevate the average. This difference is particularly pronounced in the United States, suggesting greater income dispersion within that market.

These findings highlight the strong influence of geographic location on salary outcomes. The magnitude of cross-country variation suggests that regional economic conditions and labor market structures play a central role in shaping income levels.

5.4 Job Satisfaction vs AI Usage

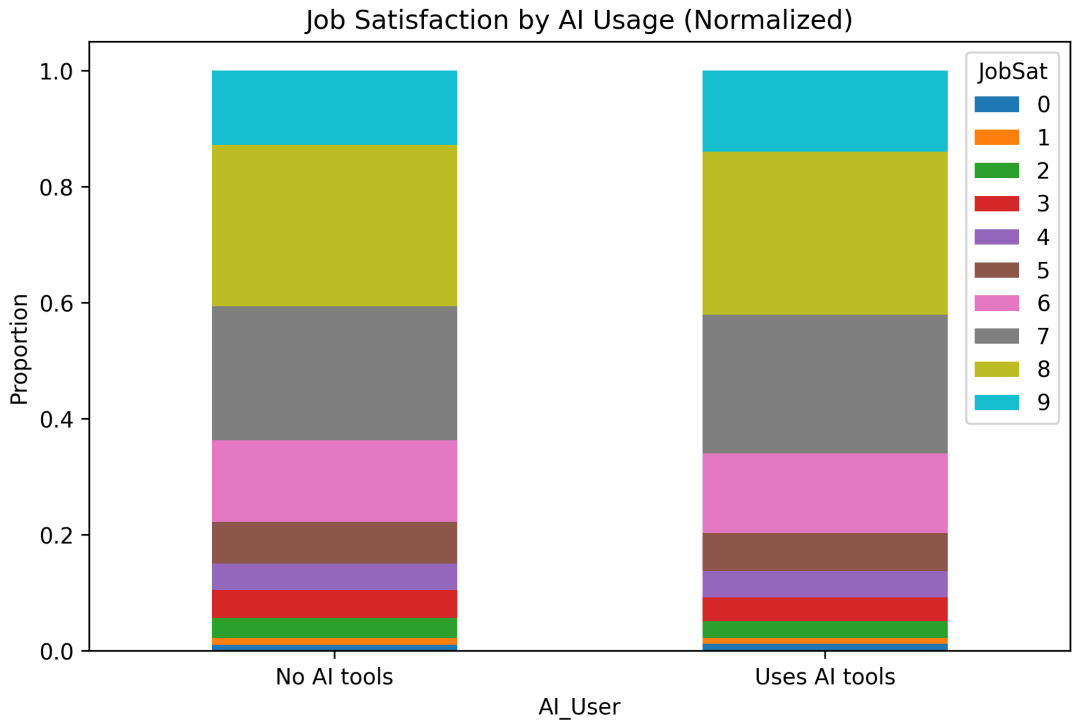

Figure 13: Job Satisfaction Distribution by AI Usage

The normalized distribution of job satisfaction levels shows broadly similar patterns between AI users and non-users. Across both groups, higher satisfaction levels (values 7–9) account for a substantial proportion of responses.

AI users exhibit a slightly greater concentration in the upper satisfaction categories, suggesting a modest association between AI tool usage and higher reported job satisfaction. However, the overall distribution remains largely comparable between the two groups.

The similarity in proportions across most satisfaction levels indicates that AI adoption is not associated with dramatic differences in job satisfaction. As with previous comparisons, this relationship reflects correlation rather than causation.

6 Correlation Analysis

6.1 Role-Specific Salary Differences and AI Usage

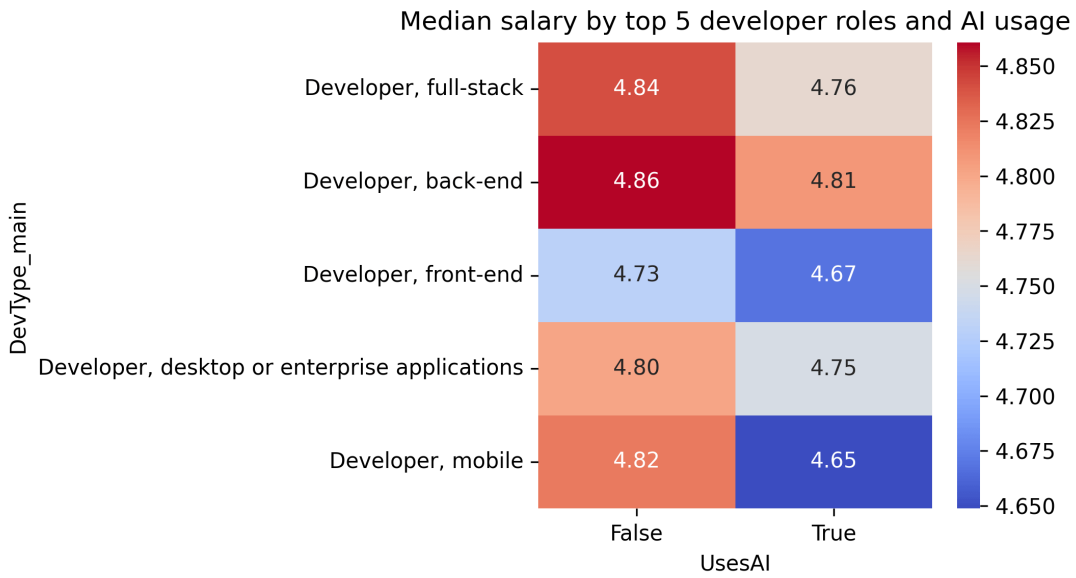

Figure 14: Median Salary (log10) by Developer Role and AI Usage

To further investigate whether AI adoption is associated with compensation differences, salary medians were examined within the top five developer roles. This stratified analysis controls for role-specific salary structures and allows for a more nuanced interpretation.

Across all listed roles, the median log-salary of AI users is slightly lower than that of non-users. While the differences are modest, they appear consistent. The gap is most pronounced among mobile developers, whereas back-end developers exhibit the smallest variation.

These findings suggest that the positive salary association observed in aggregate comparisons does not uniformly persist when controlling for professional role. In other words, AI usage alone does not appear to confer a salary premium within specific developer categories.

Overall, the results highlight the importance of stratified analysis. Aggregate comparisons may obscure underlying structural differences across roles, leading to potentially misleading conclusions about the economic impact of AI adoption.

7 Discussion

The results of this study provide a structured and multi-layered perspective on the evolving relationship between professional experience, geographic location, developer roles, and AI adoption within the global software workforce.

First, professional experience emerges as a statistically meaningful predictor of income. The weak-to-moderate positive correlation ($r = 0.325$) confirms that accumulated human capital contributes positively to salary growth. However, the substantial dispersion observed within experience groups indicates that experience alone does not fully explain compensation variability. This heterogeneity suggests that structural factors such as geographic location, specialization, and labor market segmentation play significant complementary roles.

Second, geographic location appears to be a dominant determinant of salary differences. The pronounced gap between the United States and other countries, as reflected in both median and mean salary comparisons,

underscores the influence of macroeconomic conditions, purchasing power parity, and local labor market dynamics. In many cases, cross-country disparities exceed the differences observed between AI users and non-users, highlighting the central role of regional economic structure.

Third, the association between AI adoption and salary appears limited when examined more closely. While aggregate comparisons suggest slightly higher salaries among AI users, stratified analyses controlling for developer role reveal that this pattern does not consistently persist. Within several roles, AI users exhibit similar or even marginally lower median salaries compared to non-users. This indicates the presence of composition effects and potential confounding factors, where AI usage may correlate with certain professional profiles rather than directly influencing compensation levels.

Furthermore, job satisfaction distributions between AI users and non-users remain largely comparable. Although AI tools may enhance workflow efficiency and productivity, their adoption does not appear to fundamentally transform overall job satisfaction levels. Broader workplace conditions, organizational environment, and career progression opportunities likely exert stronger influence on subjective well-being outcomes.

Taken together, these findings suggest that AI tools function primarily as productivity-enhancing instruments rather than independent economic drivers. Their impact on compensation appears contextual and mediated by experience, professional role, and geographic environment. The evidence underscores the importance of stratified and multivariate perspectives when assessing the economic implications of emerging technologies.

8 Conclusion

This study examined the 2024 Stack Overflow Developer Survey to explore the relationships between professional experience, geographic location, developer roles, AI adoption, and compensation outcomes within the global software workforce.

The findings confirm that professional experience and geographic location remain the primary determinants of salary variation. While AI adoption is widespread and associated with slight aggregate salary differences, its independent economic effect appears limited once structural factors such as role specialization and country are taken into account.

The analysis challenges oversimplified narratives suggesting that AI usage directly translates into higher earnings. Instead, AI tools appear to function primarily as productivity enhancers within existing labor market structures rather than as transformative drivers of income inequality. Stratified evidence highlights the importance of controlling for role composition and geographic context when evaluating the economic implications of technological adoption.

Future research could extend this work through longitudinal data to assess whether AI adoption generates long-term income mobility or remains primarily associated with efficiency gains. Multivariate modeling approaches may further disentangle the relative contributions of experience, specialization, and regional economic conditions.

In conclusion, while AI is increasingly embedded in professional practice, the core architecture of compensation in the developer labor market continues to be shaped predominantly by experience, specialization, and geography. Technological adoption operates within these structural boundaries rather than redefining them.